Integrallash jadvali

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1; \quad \int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + C; \quad \int \frac{dx}{x^2} = -\frac{1}{x} + C.$

2. $\int \frac{dx}{x} = \ln|x| + C.$

3. $\int a^x dx = \frac{a^x}{\ln a} + C; \quad \int e^x dx = e^x + C.$

4. $\int \cos x\, dx = \sin x + C.$

5. $\int \sin x\, dx = -\cos x + C.$

6. $\int \frac{dx}{\cos^2 x} = tgx + C.$

7. $\int \frac{dx}{\sin^2 x} = -ctgx + C.$

8. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\frac{x}{a} + C; \int \frac{dx}{\sqrt{1 - x^2}} = \arcsin x + C.$

9. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\operatorname{arc}tg\frac{x}{a} + C; \int \frac{dx}{1 + x^2} = \operatorname{arc}tgx + C.$

10. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C.$

11. $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln\left|x + \sqrt{x^2 \pm a^2}\right| + C.$

1. Bevosita integrallash usulidan foydalanib aniqmas integralni toping:

a) $\int x^6 dx = \frac{x^7}{7} + C.$

b) $\int \frac{dx}{\sqrt[3]{x}} = \int x^{-\frac{1}{3}} dx = \frac{x^{\frac{2}{3}}}{\frac{2}{3}} + C = \frac{3\sqrt[3]{x^2}}{2} + C.$

v) $\int \frac{dx}{4+x^2} = \int \frac{dx}{2^2+x^2} = \frac{1}{2} arctg \frac{x}{2} + C.$

g) $\int \left(2x^3 - 3\sin x + 5\sqrt{x}\right) dx = \int 2x^3 dx - 3\int \sin x dx + 5\int x^{\frac{1}{2}} dx =$

$= 2\frac{x^4}{4} + 3\cos x + 5\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = \frac{x^4}{2} + 3\cos x + \frac{10x\sqrt{x}}{3} + C.$

d) $\int \sin\left(x + \frac{\pi}{3}\right) dx = \int \sin\left(x + \frac{\pi}{3}\right) d\left(x + \frac{\pi}{3}\right) = -\cos\left(x + \frac{\pi}{3}\right) + C.$

2. O‘zgaruvchini almashtirish usulidan foydalanib aniqmas integralni toping: $\int \frac{dx}{\sqrt{e^x - 1}}$.

Yechish. $e^x - 1 = t^2$ almashtirish kiritamiz. U holda $e^x = t^2 + 1$, $x = \ln\left(t^2 + 1\right)$, $dx = \frac{2t}{t^2+1} dt$ va

$\int \frac{dx}{\sqrt{e^x - 1}} = \int \frac{2dt}{t^2+1} = 2arctgt + C = 2arctg\sqrt{e^x - 1} + C$ bo‘ladi. ▶

1. Bo’laklab almashtirish usulidan foydalanib aniqmas integralni toping: $\int x \cos x dx$.

Yechish. $u = x$, $du = dx$, $v = \sin x$, $dv = \cos x dx$, belgilashlarni kiritamiz. U holda

$\int x \cdot \cos x dx = x \cdot \sin x - \int \sin x dx = x \sin x + \cos x + C$ bo‘ladi. ▶

2. Firmaning marjinal daromad funksiyasi $MR=150-15Q$ ko‘rinishida berilgan. Agar $Q=10$birlik mahsulot ishlab chiqarilgan dafirmaning umumiy daromadi $TR=1000$p.b. ni tashkil etsa, u holda yalpi daromad funksiyasi qanday ko‘rinishda bo‘ladi.

Yechish. Umumiy daromad funksiyasi $TR(Q)$ni quyidagi integraldan foydalanib topamiz:

$Q=10$ da $TR(Q)=1000$bo‘lganidan foydalanib, $S=250$ekanligini aniqlaymiz. Demak, firmaning yalpi daromad funksiyasi

$TR=150Q-\frac{15Q^2}{2}+250$ ko‘rinishda bo‘ladi. ▶

3. Firmaning marjinal xarajat funksiyasi $MS=5Q^2-5Q-8$ko‘rinishga ega. Agar $Q=6$birlik mahsulot ishlab chiqarganda firmaning xarajati 500 p.b. ni tashkil etsa, u holda uning umumiy xaraja tfunksiyasini toping.

Yechish.Umumiy xarajat funksiyasi $TC(Q)$ni quyidagi integraldan foydalanib topamiz:

$$TC(Q)=\int MC(Q)\,dQ=\int(5Q^2-5Q-8)\,dQ=5\frac{Q^3}{3}-5\frac{Q^2}{2}-8Q+C.$$

$Q=6$ da $TC(Q)=500$ekanidan foydalanib, $S=278$ekanligini aniqlaymiz. Demak, umumiy xarajatlar funksiyasi quyidagi ko‘rinishda bo‘ladi: $TC(Q)=5\frac{Q^3}{3}-5\frac{Q^2}{2}-8Q+278.$ ▶

? Quyidagi integrallarni toping:

4. $\int\frac{2-\sqrt{1-x^2}}{\sqrt{1-x^2}}dx$;

5. $\int\left(x^2+2x+\frac{1}{x}\right)dx$;

6. $\int\frac{x-2}{x^3}dx$;

7. $\int\left(\sqrt{x}+\sqrt[3]{x}\right)dx$;

8. $\int\frac{\left(\sqrt{x}-1\right)^3}{x}dx$;

9. $\int\frac{x-1}{\sqrt[3]{x^2}}dx$;

10. $\int\frac{\cos 2x}{cos^2x\ sin^2x}dx$;

11. $\int ctg^2x\ dx$;

12. $\int\frac{dx}{\sin^2 x\cos^2 x}$;

? O‘zgaruvchini almashtirishusulidan foydalanib integrallarni toping:

13. $\int x^3(1-2x^4)^3dx$;

14. $\int x\sqrt{a-x}dx$, $a-x=t^2$;

15. $\int\frac{dx}{x\ln x}$;

16. $\int\frac{cosx\ dx}{\sqrt{1+2sin^2x}}$;

17. $\int\frac{\sin 2x}{\sqrt{2+cos^2x}}dx$;

18. $\int x\ cos(x^2)dx$;

? Bo‘laklab integrallash usulidan foydalanib integrallarni toping:

19. $\int\frac{\ln x}{x^3}dx$;

20. $\int x^2e^{3x}dx$;

21. $\int x\ cosxdx$;

22. $\int ln^2xdx$;

23. $\int\frac{x\cos x\ dx}{sin^3x}$;

24. $\int \frac{arcsin\frac{x}{2}}{\sqrt{2^2-x^2}}dx$;

25. $\int (x+1)e^x dx$;

26. $\int e^{2x}\cos x\, dx$;

? Quyidagi integrallarni qulay usulda toping:

27. $\int \frac{dx}{x^3}$;

28. $\int \frac{dx}{\sqrt{2-x^2}}$;

29. $\int \frac{dx}{2x^2-6}$;

30. $\int (1+e^x)^2 dx$;

31. $\int \frac{2x+3}{x^2-5}dx$;

32. $\int tg^2\varphi d\varphi$;

33. $\int \frac{2x\ dx}{x^4+3}$;

34. $\int \frac{\sin\ x\ dx}{\sqrt{1+2\cos\ x}}$;

35. $\int \frac{dx}{\sqrt{e^x+1}}$; $(e^x+1=t^2)$;

36. $\int \frac{e^x dx}{3+4e^x}$;

37. $\int e^{\sin\ x}\cos x\, dx$;

38. $\int \frac{sin\,4x}{cos^4 2x+4}dx$;

39. $\int \frac{2e^x dx}{\sqrt{16-e^{2x}}}$,

40. $\int \frac{\sqrt{2-x^2}+\sqrt{2+x^2}}{\sqrt{4-x^4}}dx$;

41. $\int x\ sinx\ dx$;

42. $\int x^3 e^x dx$;

43. $\int \sqrt{a^2-x^2}dx,\ a>0,\ u=\sqrt{a^2-x^2}$;

44. $\int sin\sqrt{x}dx$; $\sqrt{x}=t$;